

Introduction To MYSQL

SQL is built by 'Database'. To communicate the Database by "MySQL". "Oracle", etc., "MySQL" is the base language. MySQL is stored in Structured of "Rows" and "Columns" like a Table.

General Database Structure

Database Software:

- Software used to create and manage database, such as **MySql**, postgresql, and **MS SQL**

Database:

- A place where related data is stored and organized separately.

Scheme:

- A collection of table organized logically. It works like a blueprint of the database.

Tables:

- A structure that store data rows and columns such as users or product information.

Data:

- The actual information stored in the database, such names, ages, price and dates.

There are different types of MySQL:

DDL (Data definition language)

Purpose: Defines, modifies, and alters the actual structure or schema of the database.

Core Commands:

- INSERT (adds new rows)
- UPDATE (edits existing data)
- DELETE (removes specific records).

TCL (Transaction Control Language)

Purpose: Manages transactions to group multiple tasks together, ensuring database integrity.

Core commands:

- COMMIT (saves changes permanently)
- ROLLBACK (undoes changes if an error occurs)

Datatypes of SQL:

NUMERIC-----	DATE AND TIME-----	STRING
int	date and time("Y:M:D")	varchar
bigint	timestamp (change in universal time)	
Float		

Movie Database Management and Analytical Reporting Case Study:

#Create a database and use the database

```

create database Movies_DB;

use Movies_DB;

create table movie
(
rel_year int,
mov_name varchar(100),
act_name varchar(50),
lang varchar(30),
dir_name varchar(100)
);

```

Insert the data's

```

INSERT INTO MOVIES VALUES (1960,'Kalathur Kannamma','Kamal Haasan','Tamil','A. Bhimsingh');
INSERT INTO MOVIES VALUES (1962,'Kannum Karalum','Kamal Haasan','Malayalam','K. S. Sethumadhavan');
INSERT INTO MOVIES VALUES (1962,'Paadha Kaanikkai','Kamal Haasan','Tamil','K. Shankar');
INSERT INTO MOVIES VALUES (1962,'Paarthaal Pasi Theerum','Kamal Haasan','Tamil','A. Bhimsingh');
INSERT INTO MOVIES VALUES (1963,'Anandha Jodhi','Kamal Haasan','Tamil','V. N. Reddy');
INSERT INTO MOVIES VALUES (1963,'Vanambadi','Kamal Haasan','Tamil','G. R. Nathan');
INSERT INTO MOVIES VALUES (1970,'Maanavan','Kamal Haasan','Tamil','M. A. Thirumugam');
INSERT INTO MOVIES VALUES (1971,'Annai Velankanni','Kamal Haasan','Tamil','Thankappan');
INSERT INTO MOVIES VALUES (1971,'Anubhavangal Paalichakal','Mammootty','Malayalam','K. S. Sethumadhavan');
INSERT INTO MOVIES VALUES (1972,'Kurathi Magan','Kamal Haasan','Tamil','K. S. Gopalakrishnan');
INSERT INTO MOVIES VALUES (1973,'Arangetram','Kamal Haasan','Tamil','K. Balachander');
INSERT INTO MOVIES VALUES (1973,'Kaalachakram','Mammootty','Malayalam','K. Narayanan');
INSERT INTO MOVIES VALUES (1973,'Sollathaana Ninaikkiren','Kamal Haasan','Tamil','K. Balachander');
INSERT INTO MOVIES VALUES (1974,'Anbu Thangai','Kamal Haasan','Tamil','S. P. Muthuraman');
INSERT INTO MOVIES VALUES (1974,'Aval Oru Thodar Kathai','Kamal Haasan','Tamil','K. Balachander');

```

1. Retrieve all details from the movie table.

```
select * from MOVIES;
```

Solution:

rel_year	mov_name	act_name	lang	dir_name
1960	Kalathur Kannamma	Kamal Haasan	Tamil	A. Bhimsingh
1962	Kannum Karalum	Kamal Haasan	Malayalam	K. S. Sethumadhavan
1962	Paadha Kaanikkai	Kamal Haasan	Tamil	K. Shankar
1962	Paarthaal Pasi Theerum	Kamal Haasan	Tamil	A. Bhimsingh
1963	Anandha Jodhi	Kamal Haasan	Tamil	V. N. Reddy
1963	Vanambadi	Kamal Haasan	Tamil	G. R. Nathan
1970	Maanavan	Kamal Haasan	Tamil	M. A. Thirumugam
1971	Annai Velankanni	Kamal Haasan	Tamil	Thankappan
1971	Anubhavgal Paalichakal	Mammootty	Malayalam	K. S. Sethumadhavan
1972	Kurathi Magan	Kamal Haasan	Tamil	K. S. Gopalakrishnan
1973	Arangetram	Kamal Haasan	Tamil	K. Balachander
1973	Kaalachakram	Mammootty	Malayalam	K. Narayanan
1973	Sollathaana Ninaikkiren	Kamal Haasan	Tamil	K. Balachander
1974	Anbu Thangai	Kamal Haasan	Tamil	S. P. Muthuraman
1974	Aval Oru Thodar Kathai	Kamal Haasan	Tamil	K. Balachander

2 . Display only movie names and actor names.

```
SELECT mov_name, act_name
FROM movies;
```

Solution:

	mov_name	act_name
►	Kalathur Kannamma	Kamal Haasan
	Kannum Karalum	Kamal Haasan
	Paadha Kaanikkai	Kamal Haasan
	Paarthaal Pasi Theerum	Kamal Haasan
	Anandha Jodhi	Kamal Haasan
	Vanambadi	Kamal Haasan
	Maanavan	Kamal Haasan
	Annai Velankanni	Kamal Haasan
	Anubhavangal Paalichakal	Mammootty
	Kurathi Magan	Kamal Haasan
	Arangetram	Kamal Haasan
	Kaalachakram	Mammootty
	Sollathan Ninaikkiren	Kamal Haasan
	Anbu Thangai	Kamal Haasan
	Aval Oru Thodar Kathai	Kamal Haasan
	Gumasthavin Magal	Kamal Haasan
	Kanyakumari	Kamal Haasan

3.Display distinct movie languages.

```
select distinct lang from movies;
```

Solution:

Result Grid				File
	lang			
▶	Tamil			
	Malayalam			
	Telugu			
	Kannada			
	Hindi			
	Bengali			
	Silent			
	English			

4.Retrieve unique actor names.

```
SELECT DISTINCT act_name  
FROM movies;
```

Solution:

	act_name
►	Kamal Haasan
	Mammootty
	Rajinikanth
	Mohanlal
	Vijay
	Silambarasan
	Ajith Kumar
	Suriya
	Trisha
	Nayanthara
	Karthi
	Dhanush
	Dulquer Salm...
	Amala Paul
	Rashmika Ma...
	Manju Warior
	Sai Pallavi
	Parvathy Thi...
	Vijay Sethup...